

CPEN 416

Lecture 03: Single-qubit operations and projective measurement

Thursday 17 September 2026

Announcements

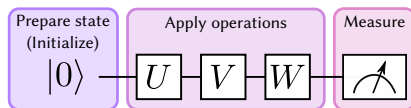
- Practice quiz / assignment questions becoming available
- Tutorial 2 on Monday
- Quiz 2 on Tuesday (based on content from lectures 2 and 3)

back in person

Last time

We represented quantum computations in two new ways.

Quantum circuits:



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

observe qubit
in $|0\rangle$ w/prob $|\alpha|^2$
in $|1\rangle$ w/prob $|\beta|^2$

$$|\alpha|^2 + |\beta|^2 = 1$$

Quantum functions (and QNodes) in PennyLane:

```
import pennylane as qnp

dev = qnp.device("default.qubit", wires=1)

@qnp.qnode(dev)
def my_quantum_function():
    qnp.Hadamard(wires=0)
    qnp.PauliZ(wires=0)
    return qnp.probs()
```

Last time

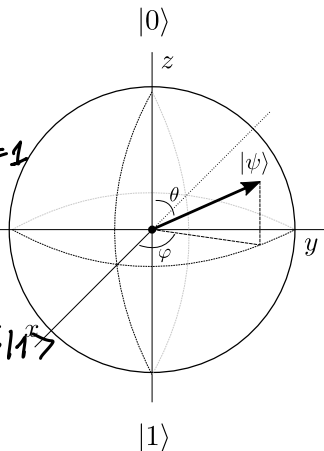
We parametrized single-qubit states with 2 real numbers, and plotted states in 3D space on the Bloch sphere.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

4 params initially
-1 for norm. $|\alpha|^2 + |\beta|^2 = 1$
-1 global phase

2 parameters

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle$$

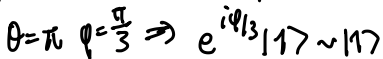


<https://bloch.kherb.io/>

Learning outcomes

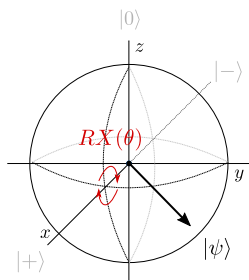
- 1 Describe the behaviour of common single-qubit gates
- 2 Compute the inner product between two quantum states
- 3 Perform a projective measurement in the computational basis
- 4 Apply basis rotations to perform a projective measurement in any basis

"antipodal"
(opposite poles)



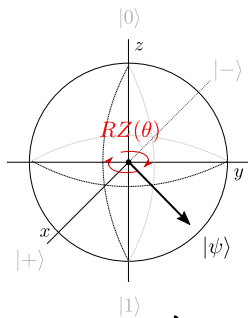
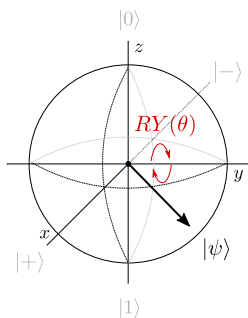
Rotations: the Bloch sphere

Unitary operations rotate the state vector on the Bloch sphere.



$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

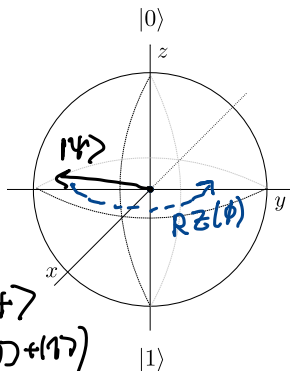
Image credit: Codebook node 1.6



$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Z rotations (RZ)

$$RZ(\phi) = e^{-i\frac{\phi}{2}Z} = \begin{pmatrix} e^{-i\frac{\phi}{2}} & 0 \\ 0 & e^{i\frac{\phi}{2}} \end{pmatrix}$$



In PennyLane:

```
qp.RZ(phi, wires=wire)
```

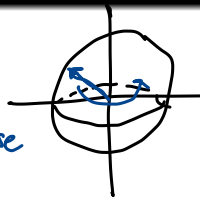
Try at home: directly expand $e^{-i\frac{\phi}{2}Z}$ to obtain the matrix representation.

Z rotations (RZ)

$$RZ(\phi) |0\rangle = e^{-i\phi/2} |0\rangle \rightarrow \text{eigenvectors!}$$

$$RZ(\phi) |1\rangle = e^{i\phi/2} |1\rangle \quad RZ(\phi) = e^{-i\frac{\phi}{2}Z} = \begin{pmatrix} e^{-i\frac{\phi}{2}} & 0 \\ 0 & e^{i\frac{\phi}{2}} \end{pmatrix}$$

relative phase



Apply to a general state:

$$|\psi\rangle = \cos\frac{\theta}{2} |0\rangle + e^{i\varphi} \sin\frac{\theta}{2} |1\rangle$$

$$RZ(\phi) |\psi\rangle = \cos\frac{\theta}{2} RZ(\phi) |0\rangle + e^{i\varphi} \sin\frac{\theta}{2} RZ(\phi) |1\rangle$$

θ : expect unchanged
 φ : will change

$$= \cos\frac{\theta}{2} e^{-i\phi/2} |0\rangle + e^{i\varphi} \sin\frac{\theta}{2} e^{i\phi/2} |1\rangle$$

$$= e^{-i\phi/2} \left[\cos\frac{\theta}{2} |0\rangle + e^{i\varphi} \sin\frac{\theta}{2} e^{i\phi} |1\rangle \right]$$

$$RZ(\phi) = \begin{pmatrix} e^{-i\phi/2} & 0 \\ 0 & e^{i\phi/2} \end{pmatrix}$$

$$= e^{-i\phi/2} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$$

global phase

$$\sim \cos\frac{\theta}{2} |0\rangle + e^{i(\varphi+\phi)} \sin\frac{\theta}{2} |1\rangle$$

relative phase changes due to rotation

S and T

Two other special cases: $\phi = \pi/2$, and $\phi = \pi/4$.

$$S = RZ(\pi/2) = \begin{pmatrix} e^{-i\frac{\pi}{4}} & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

$$T = RZ(\pi/4) = \begin{pmatrix} e^{-i\frac{\pi}{8}} & 0 \\ 0 & e^{i\frac{\pi}{8}} \end{pmatrix} \sim \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{pmatrix}$$



In PennyLane:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

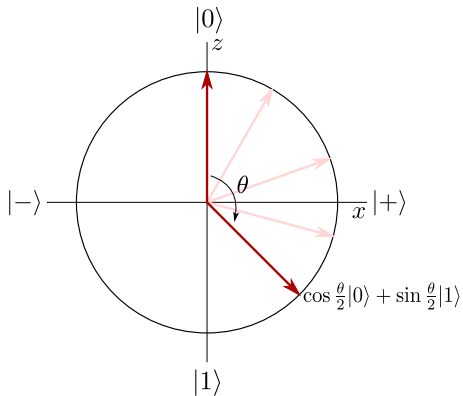
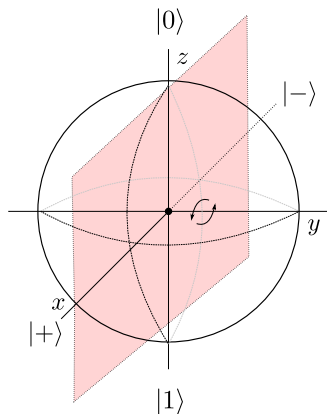
```
qp.PauliZ(wires=wire)
qp.S(wires=wire)
qp.T(wires=wire)
```

$$U U^\dagger = I$$

S is part of a special group called the **Clifford group**.

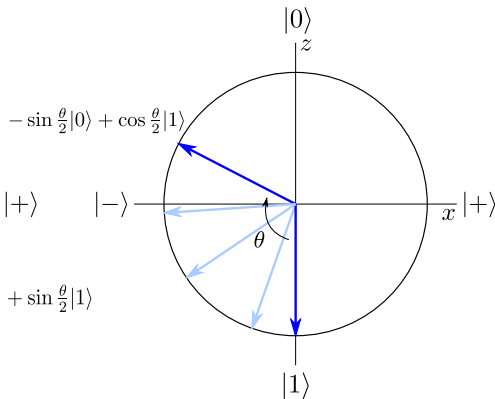
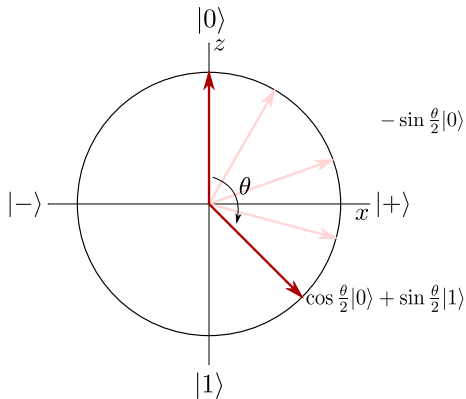
T is used in universal gate sets for fault-tolerant QC.

Y rotations (RY)



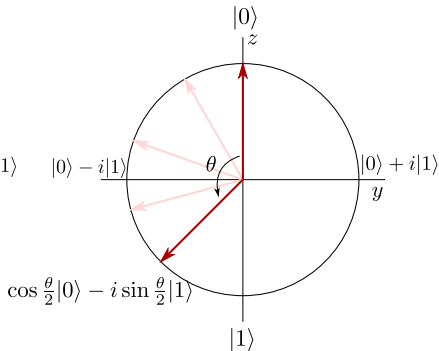
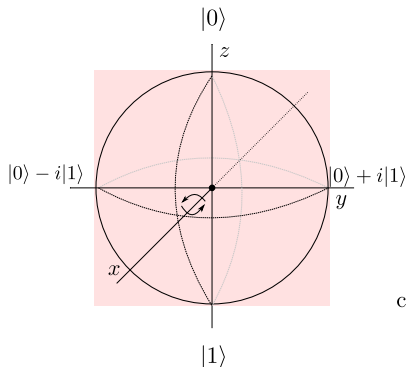
Y rotations (RY)

$$RY(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$



X rotations (RX)

$$RX(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$



Pauli rotations

These unitary operations are called **Pauli rotations**.

	Math	Matrix	Code	Special cases (θ)
RZ	$e^{-i\frac{\theta}{2}Z}$	$\begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$	<code>qp.RZ</code>	$Z(\pi), S(\pi/2), T(\pi/4)$
RY	$e^{-i\frac{\theta}{2}Y}$	$\begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$	<code>qp.RY</code>	$Y(\pi)$
RX	$e^{-i\frac{\theta}{2}X}$	$\begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$	<code>qp.RX</code>	$X(\pi), SX(\pi/2)$

Pauli rotations

Exercise: design a quantum circuit that prepares

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle - \frac{1}{2}e^{i\frac{5\pi}{4}}|1\rangle$$

Hint: you can return the state directly in PennyLane:

```
@qp.qnode(dev)
def some_circuit():
    # Gates...
    return qp.state()
```

Pauli rotations

Exercise: implement the circuit below in PennyLane.

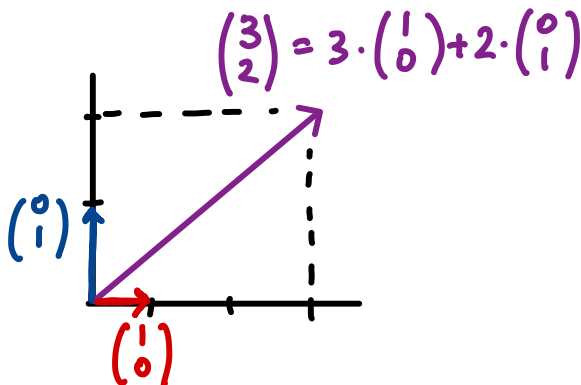


Run your circuit with two different values of θ and take 1000 shots.

How does θ affect the measurement outcome probabilities?

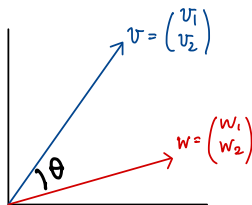
Inner products

We can now create any single-qubit quantum state: how do we *compare* them? Look to 2D space for intuition.



Inner products

The **inner product** (dot product) between two vectors quantifies how much overlap they have.



$$\begin{aligned} \langle \vec{v}, \vec{w} \rangle &= \vec{v}^T \cdot \vec{w} = \begin{pmatrix} v_1 & v_2 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = v_1 w_1 + v_2 w_2 \\ &= |\vec{v}| |\vec{w}| \cos \theta \end{aligned}$$

Inner products

Take just one of these representations:

$$\langle \vec{v}, \vec{w} \rangle = \vec{v}^T \vec{w}$$

$$(|v\rangle)^\dagger = (|v\rangle^T)^*$$

The inner product in Hilbert space is defined as

$$|v\rangle = v_1|0\rangle + v_2|1\rangle \quad v_i, w_i \in \mathbb{C}$$

$$|w\rangle = w_1|0\rangle + w_2|1\rangle$$

$$\Rightarrow \langle |v\rangle, |w\rangle \rangle = (v_1^* \ v_2^*) \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = (|v\rangle)^\dagger |w\rangle$$

To avoid cumbersome notation define the **bra** (*braket* notation):

$$\langle v| = (|v\rangle)^\dagger = (v_1^* \ v_2^*)$$

$$\langle v| w \rangle = v_1^* w_1 + v_2^* w_2$$

Inner products

Exercise: compute the inner product of the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

with itself.

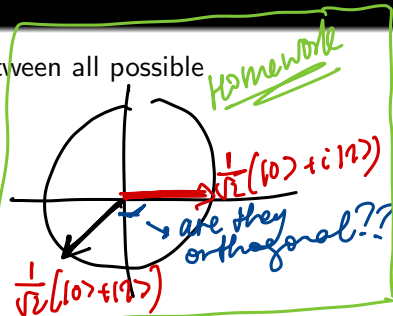
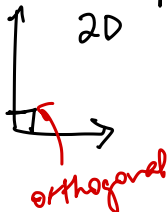
$$\begin{aligned} \langle\psi|\psi\rangle &= (\alpha^* \ \beta^*) \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \\ &= \alpha^* \alpha + \beta^* \beta \\ &= 1 \end{aligned}$$

Inner products

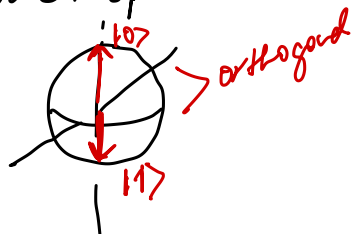
Exercise: compute the inner product between all possible combinations of $|0\rangle$ and $|1\rangle$.

$\langle 0 0\rangle$	1
$\langle 0 1\rangle$	0
$\langle 1 0\rangle$	0
$\langle 1 1\rangle$	1

graphical repr:



Bloch sphere



Next time

Content: *projective meas. and basis rotations*

- Mathematical representation of multi-qubit systems
- Multi-qubit gates
- Entanglement

Action items:

- ① Prepare for quiz 2 (tutorial on Monday)
- ② Work on practice Assignment 1

Recommended reading:

- For today: Codebook modules IQC, SQ; Nielsen & Chuang 1.3.3, 2.2.3-2.2.5
- For next class: Codebook module MQ; Nielsen & Chuang 4.3